

BIOLOGY BOARD QUESTION PAPER : JULY 2024**Time: 3 Hrs. Max. Marks: 70****General Instructions:**

(1) The question paper is divided into four sections.

Section A: Q.1 contains Ten multiple choice type questions of one mark each. Evaluation will be done for the first attempt only.

Section A: Q.2 contains Eight very short answer type questions of one mark each.

(2) Section B: Q.3 to Q.10 are short answer type questions carrying two marks each.
(Attempt any Eight)

(3) Section C: Q.11 to Q.26 are long answer type questions carrying three marks each.
(Attempt any Eight)

(4) Section D: Q.27 to Q.31 are long answer type questions carrying four marks each.
(Attempt any Three)

(5) Begin each section on a new page.

SECTION – A

Q.1 Select and write the correct answer for the following multiple choice type questions: [10]

i. In monocot embryo the protective sheath of plumule is called—

(A) perisperm (B) scutellum (C) coleorhiza (D) coleoptile

ii. The synonymous term for telomere is—

(A) primary constriction (B) telomere (C) secondary constriction (D) satellite

iii. Protein digesting enzymes are—

(A) polymerase (B) protease (C) pectinase (D) RNAase

iv. Spermatids get transformed into functional spermatozoa by—

(A) spermiogenesis (B) spermiation (C) gametogenesis (D) oogenesis

v. Industrial melanism is example of—

(A) natural selection (B) artificial selection (C) reproductive isolation
(D) adaptive variation

vi. Value of root pressure is +10 to +20 bars which lifts water to—

(A) 5–60 m (B) 10–120 m (C) 30–40 m (D) 20–30 m

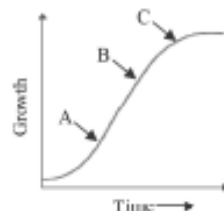
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vii. Cellular fragments formed from megakaryocytes are—
(A) erythrocytes (B) leucocytes (C) thrombocytes (D) monocytes

viii. Dopamine neuron degeneration causes—
(A) Alzheimer's (B) Addison's (C) Grave's (D) Parkinson's disease

ix. Correct order of phases A, B, C in fig—
(A) Stationary, Log, Lag (B) Log, Lag, Stationary
(C) Log, Stationary, Lag (D) Lag, Log, Stationary



x. River Popper Hypothesis explains—
(A) loss of biodiversity (B) biomagnification (C) species–area relationship
(D) significance of diversity

Q.2 Answer the following questions: [8]

- Ratio of methane, ammonia, hydrogen in Urey–Miller experiment.
- Gene introduced in tobacco against *Zabrotes subfasciatus*.
- Haploid nuclei in mature female gametophyte of angiosperms.
- Connection between pulmonary trunk and aorta (embryonic ductus arteriosus).
- Endocrine gland responsible for immune system development.
- Type of association between sea anemone and clown fish.
- Pioneer species in aquatic habitat during primary succession.

SECTION – B (Attempt any EIGHT) [16]

- Q.3 i. Why DNA replication is semi-conservative? ii. Scientists who proved it.
- Q.4 Distinguish Asexual vs Sexual reproduction w.r.t i. Cell division ii. Genetic makeup
- Q.5 Arrange in increasing cranial capacity: i. Homo sapiens ii. Ramapithecus iii. Australopithecus iv. Homo erectus
- Q.6 Identify disorder due to non-disjunction of 21st chromosome + characteristics.
- Q.7 What is radial & tangential translocation of food in plants?
- Q.8 Any two causes of genetic variation.
- Q.9 Identify A, B, C, D:

Organism	Habitat	Respiratory organ
A	Terrestrial	Tracheal tubes + spiracles
Scorpions	Terrestrial	B
Frogs	C	Plasma membrane
Turtles	Underwater	D



Q.10 Function of Nabenken and Zona pellucida.

Q.11 Location of corpus callosum & arbor vitae.

Q.12 Match the following:

Column-I	Column-II
1. Cytotoxic T-cells	(a) Activation of helper
2. Plasma cells	(b) Blood group antigens
3. Antigen presenting cells	T-cells(c) Secretion of perforins
4. Red blood cells	(d) Antibody production

SECTION – C

(Attempt any EIGHT) [24]

Q.13 Any two commercially used products of transgenic plants.

Q.14 Causative agent, symptoms, mode of transmission of typhoid.

Q.15 Terms: i. Male & female reproductive organs in same organism ii. Y-linked gene iii. Haploid male bee.

Q.16 Role of diacylglycerols in inflammation.

Q.17 Cardiac cycle: i. atrial systole ii. ventricular systole iii. joint diastole

Q.18 Reflex action and two examples based on experience.

Q.19 Fill table with: Sickle cell anaemia, Widow's peak, Flower colour of Mirabilis, Human blood groups, Coat colour of cattle, Klinefelter's syndrome.

Genetic trait	Example
i. Incomplete dominance	
ii. Pleiotropy	
iii. Co-dominance	
iv. Multiple allelism	
v. Sex chromosomal disorder	
vi. Autosomal disorder	

Q.20 Role of: Phyllosphere, Allelmaria, Flavobacterium

Q.21 Regulators, Conformers, Migrate

Q.22 Any three causes of loss of biodiversity.

Q.23 Physiological effects & uses of cytokinin.

Q.24 Mycorrhiza: types & benefits.

Q.25 Explain the following

- i. Leaching ii. Humification & mineralisation
- iii. Humus iv. Homeostasis v. Temporary ↑WBCs
- vi. Decrease in RBCs



SECTION – D**(Attempt any THREE) [12]**

Q.27 Stomata opening & closing: starch–sugar theory; proton transport theory.

Q.28 i. Hormone definition

ii. Two properties

iii. Diagram of membrane-bound receptor hormone action.

Q.29 Diagram-based: i. Type of ovule

ii. First cell of embryo sac

iii. Identify A & B and their functions.

Q.30 Commercial benefits of transgenic pigs & cattle.

Q.31 Histology of testis with diagram.

